

Agentic Epistemology: Facet-Conditioned Auditability for Autonomous Agents and Synthetic Societies

Jason Roell

January 15, 2025

Abstract

Autonomous agents and synthetic personas increasingly act from long-lived context rather than single-turn instructions. Their failures are therefore epistemic: the system may overweight the wrong evidence, ignore a persona constraint, express confidence unsupported by its latent state, or provide a plausible but unfaithful justification. This paper introduces the Agentic Epistemology Framework (AEF), an audit architecture for representing and tracing beliefs, confidence, justifications, frames, and facet-conditioned persona constraints. The framework combines structured observer traces with black-box faithfulness tests, confidence calibration checks, and optional representation-level probes inspired by recent work on chain-of-thought faithfulness, model self-knowledge, representation engineering, and sparse autoencoder interpretability. We extend the framework with *facets*: persistent psychological, socio-ecological, psychographic, and interaction constraints that compose nonlinearly into an agent’s active frame. A TypeScript reference implementation includes automated benchmarks for root-cause diagnosis and facet-frame faithfulness. In controlled synthetic cases, full epistemic traces recover injected generic root causes with 1.000 accuracy and recover injected facet-faithfulness failures with 0.665 accuracy, while ablated traces degrade substantially. These results support the central claim that explicit epistemic state and facet-aware traces make agent behavior more diagnosable, while leaving open the harder problem of validating open-ended real-world simulations.

1 Introduction

Autonomous agents now plan, use tools, communicate, and revise internal state across long-running interactions. Synthetic personas add another layer: they must remain faithful to a constructed identity while responding naturally to novel stimuli. In both cases, important causal factors are often hidden inside prompt text, model activations, unstructured logs, or generated explanations that may not faithfully describe why the model produced a response.

This paper introduces the Agentic Epistemology Framework (AEF), a structured architecture for auditing agent reasoning. AEF represents beliefs as propositions with confidence and justifications, records observer events for epistemic transitions, and models frames as active cognitive perspectives that influence evidence weighting, source trust, and action thresholds. The paper further extends frames with *facets*: typed persona constraints that jointly shape the active frame. Facets can represent psychological dispositions, lived-context constraints, psychographic motivations, and interaction style. In a synthetic human simulation, facets are not independent prompt fragments. They interact nonlinearly, and those interactions should be inspectable.

The first publication of AEF has two goals. First, it defines a vocabulary and implementation pattern for epistemic auditability in agents and synthetic societies. Second, it provides executable benchmarks that test whether the resulting traces make known failure causes diagnosable. The framework does not claim that generated persona behavior is valid in the world merely because the system produces plausible text. Instead, it asks whether the system can expose enough structured evidence to debug when a persona fails to follow its own construction.

1.1 Contributions

This paper makes four contributions:

1. A structured epistemic ontology for beliefs, confidence, justifications, frames, facets, perceptions, actions, and observer events.
2. A facet composition model in which persistent persona constraints contribute nonlinearly to an agent’s active frame.
3. A TypeScript reference implementation with automated root-cause and facet-faithfulness benchmarks.
4. An evaluation protocol that combines structured traces with confidence calibration, black-box perturbation tests, and optional representation-level probes for reasoning faithfulness.

2 Background and Motivation

AEF draws from belief-desire-intention models [Bratman, 1987, Rao and Georgeff, 1991, Wooldridge, 2009], epistemic logic [Fagin et al., 2003], belief revision [Alchourrón et al., 1985, Gärdenfors, 1988], frame semantics [Minsky, 1974, Fillmore, 1976, Lakoff, 1987], argumentation theory [Dung, 1995, Nute, 1994], and agent-based modeling [Epstein and Axtell, 1996]. These traditions provide useful structure, but current LLM-based agents create additional audit problems.

Recent work shows that natural-language explanations should not be treated as automatically faithful. Chain-of-thought explanations can systematically misrepresent the true reasons for a model’s answer [Turpin et al., 2023], and faithfulness requires intervention-style tests rather than surface plausibility alone [Lanham et al., 2023]. Calibration and self-evaluation research shows that language models can sometimes estimate whether their own answers are correct, but this depends strongly on elicitation format and task setting [Kadavath et al., 2022]. Black-box hallucination detection methods, such as sampling-based consistency checks, provide another way to test whether outputs are stable under repeated generation [Manakul et al., 2023]. Meanwhile, representation engineering and sparse autoencoder work provide tools for probing latent directions or interpretable features when model activations are available [Zou et al., 2023, Cunningham et al., 2023].

AEF treats these lines of work as complementary. Observer traces expose the explicit state the system is designed to use. Black-box perturbations test whether outputs change when facets, evidence, or prompts are altered. Calibration tests ask whether the system knows when its response is likely faithful. Representation-level probes, when available, can test whether latent activations align with the asserted frame or facet. The framework is designed to hold all of these evidence types in a single audit trail.

3 Facet-Conditioned Synthetic Personas

Synthetic human simulation treats a persona as a complex adaptive system rather than a flat list of demographic labels. A persona has internal psychological dispositions, lived-context constraints, motivations, trust heuristics, resource constraints, and linguistic interaction style. In the Vurvey implementation inspected for this work, these variables are represented as a graph of PM facets. A facet is an axis of variation such as income, response style, source trust, or domain expertise. A facet value contains stable identifiers, prompts, taxonomy metadata, and reliability fields. Personas link to selected values through persona fragments. Population molds define facet distributions, mandatory and preferred values, weights, selection modes, and constraints.

This implementation suggests a natural extension to AEF. A **facet** is a typed, persistent constraint with fields:

$$Facet_i = (code_i, value_i, domain_i, prompt_i, weight_i, confidence_i, provenance_i). \quad (1)$$

The active frame is composed from the current stimulus x_t , the persona state S_t , and selected facets $Facet_{1:n}$:

$$F_t = g(S_t, x_t, Facet_{1:n}, I_{1:n}), \quad (2)$$

where $I_{1:n}$ denotes interaction terms between facets. This form is intentionally nonlinear. For example, high extraversion can produce very different behavior when combined with fragile self-confidence and extreme time poverty than when combined with high perceived control and abundant resources.

3.1 Facet Domains

The framework organizes facets into four audit domains:

1. **Psychological facets:** stable traits, anxiety level, self-confidence archetype, and emotional regulation constraints.
2. **Socio-ecological facets:** locale, life stage, socioeconomic status, time poverty, family structure, and other lived-context constraints.
3. **Psychographic facets:** motivations, adoption-curve position, source trust, perceived behavioral control, and decision heuristics.
4. **Interaction facets:** response style, conversational friction, objection tendency, and linguistic modulation.

Each facet should be auditable at three levels: whether it was selected, whether its prompt or structured representation reached the generation path, and whether the final response reflects its expected influence.

4 Epistemic State and Operational Rules

A **belief** is a proposition P held by an agent with confidence $conf \in [0, 1]$ and a justification. A **justification** is evidence, reasoning trace, source, or derivation path supporting a belief or response. A **frame** is the active cognitive lens that modulates interpretation and evidence weighting. A **facet** is a persistent constraint that contributes to frame composition. An **observer event** is a structured record of an epistemic transition.

AEF is guided by five operational rules:

1. **Traceability:** important actions and responses should trace back to beliefs, justifications, frames, and contributing facets.
2. **Explicit uncertainty:** beliefs and evaluator judgments should expose confidence rather than binary assertions.
3. **Frame awareness:** agents should record when frames change evidence weighting, source trust, or interpretation.
4. **Facet faithfulness:** persona responses should expose which facets were expected to constrain the response and which facets were actually expressed.
5. **Intervention readiness:** traces should support counterfactual tests such as removing a facet, swapping a facet value, or perturbing evidence source trust.

5 Confidence, Frame, and Facet Updates

Let $conf_{old}$ denote an agent’s current confidence in proposition P , and let e denote new evidence relevant to P . AEF uses update functions of the form:

$$conf_{new} = f(conf_{old}, just_{old}, e, F_t). \quad (3)$$

For frame-weighted evidence, evidence e suggests a target confidence $C(e, P) \in [0, 1]$. The active frame assigns evidence weight $w_{F_t}(e)$:

$$conf_{new} = (1 - w_{F_t}(e)) \cdot conf_{old} + w_{F_t}(e) \cdot C(e, P). \quad (4)$$

For facet-conditioned personas, $w_{F_t}(e)$ can be decomposed into facet contributions:

$$w_{F_t}(e) = \sigma \left(b + \sum_i \alpha_i h(Facet_i, e) + \sum_{i < j} \beta_{ij} h(Facet_i, Facet_j, e) \right), \quad (5)$$

where h maps facets and evidence to contribution scores, α_i are individual facet weights, β_{ij} are interaction weights, and σ bounds the resulting saliency. The observer should log individual contributions and interaction terms whenever the implementation can expose them.

6 Auditing Faithfulness

AEF distinguishes four kinds of faithfulness evidence.

Trace faithfulness asks whether the system recorded the inputs and transformations it was designed to use. For synthetic personas, this includes selected facets, facet values, prompt fragments, weights, constraints, and response markers.

Behavioral faithfulness asks whether the output reflects the selected persona constraints. This can be measured by entailment checks, contradiction checks, response-style classifiers, and facet-marker coverage.

Counterfactual faithfulness asks whether changing an input that should matter changes the output in the expected direction. Examples include swapping financial outlook from crisis flow to abundance, changing response style from declarer to verbatim repeater, or replacing peer-trust with institutional-trust.

Latent faithfulness asks whether hidden representations align with the claimed frame or facet when activations are available. Representation engineering can test directions corresponding to persona-relevant concepts, and sparse autoencoders can expose interpretable features. These probes are optional because many production systems expose only black-box access, but the framework leaves room for them as stronger evidence.

7 Reference Implementation

The reference implementation is written in TypeScript and is publicly available at <https://github.com/jroell/agent-epistemology-framework>. It implements beliefs, justifications, frames, agents, memory, actions, tools, and observer events. The implementation now includes two benchmark harnesses: a generic root-cause benchmark and a facet-frame faithfulness benchmark.

The Vurvy production codebase informed the facet extension. Its PM facet graph includes global facets and versioned facet values, ‘headerPrompt’ and ‘detailPrompt’ fields used to build persona prompts, ‘PmPersonaFragment’ rows that link personas to selected facet values, mold facet configurations with weights and options, and auto-configuration services that recommend and weight facet distributions for population molds. These implementation details motivate the audit requirement: a persona response should be diagnosable back to the facet graph, prompt fragments, selected values, and composed frame that produced it.

8 Automated Root-Cause Benchmark

The generic benchmark asks whether structured epistemic traces make it possible to identify the cause of agent failures in scenarios where the ground-truth cause is known. It covers four injected causes: frame misweighting, source-trust error, threshold miscalibration, and justification gap. It evaluates four trace conditions: full trace, no-frame trace, no-justification trace, and event-only trace.

We ran the benchmark with 160 synthetic cases using seed 42. The benchmark command was:

```
npx ts-node --transpile-only examples/root-cause-benchmark.ts --seed=42 --cases=160
```

Table 1: Generic root-cause diagnosis results. Full traces recover all injected causes in this controlled benchmark. Ablating frame fields, justification/source fields, or epistemic details removes diagnostic capability.

Trace condition	Accuracy	Macro F1	Coverage	Evidence inspected	Trace events
full	1.000	1.000	1.000	8.00	4.00
no_frame	0.750	0.750	0.750	10.50	4.00
no_justification	0.500	0.500	0.500	13.25	4.00
event_only	0.000	0.000	0.000	16.00	4.00

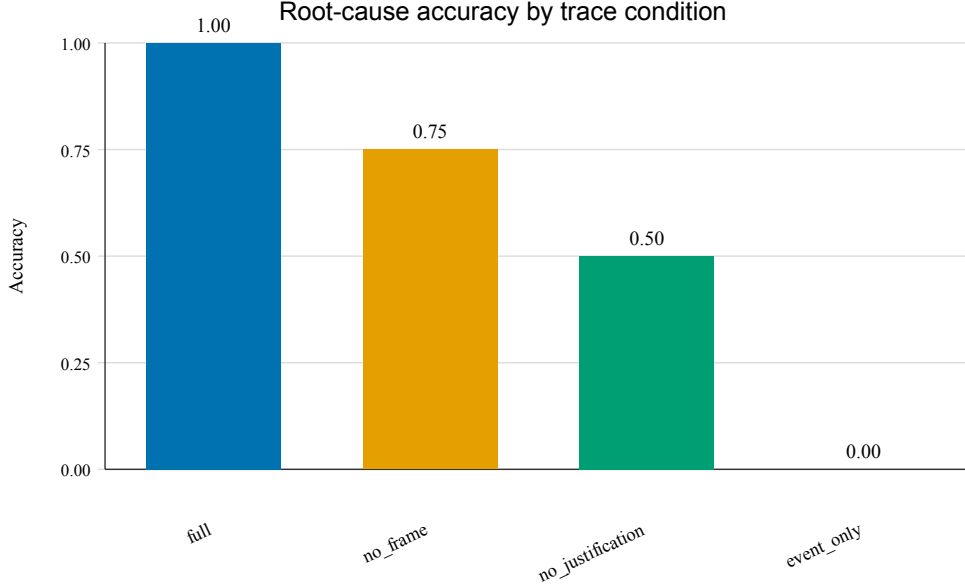


Figure 1: Generic root-cause diagnosis accuracy by trace condition.

9 Facet-Frame Faithfulness Benchmark

The second benchmark targets synthetic persona auditing. Each case constructs a persona from six facets: anxiety level, time poverty, financial outlook, source trust, response style, and adoption-curve position. The harness injects one of five failure types:

1. **Facet omission:** a required facet is absent from the composed frame.
2. **Facet misweighting:** a facet contribution is materially lower or higher than its expected calibration.
3. **Facet conflict:** two facets conflict and the response follows one while ignoring the other.
4. **Persona drift:** the response introduces an identity inconsistent with the facet set.
5. **Response-style violation:** the response ignores the selected interaction-style facet.

The benchmark evaluates four conditions: full trace, no facet weights, no facet prompts, and event-only trace. We ran 200 cases with seed 42:

```
npx ts-node --transpile-only examples/facet-frame-faithfulness-benchmark.ts --seed=42 --cases=200
```

Table 2: Facet-frame faithfulness results. Full traces outperform ablations, but the task is intentionally harder than the generic benchmark. Removing facet weights harms diagnosis of weighting errors. Removing facet prompts and expected response markers sharply reduces coverage.

Trace condition	Accuracy	Macro F1	Coverage	Events inspected	Trace events
full	0.665	0.680	0.725	24.80	8.00
no_facet_weights	0.465	0.466	0.595	29.00	8.00
no_facet_prompts	0.200	0.200	0.200	27.80	8.00
event_only	0.000	0.000	0.000	32.00	8.00

A representative facet-misweighting case illustrates the diagnostic path. The persona was expected to evaluate a premium AI subscription while constrained by crisis-flow financial outlook and a time-poverty facet. The trace showed ‘time_poverty’ with observed weight 0.18 against expected weight 0.78. The response gave a long, careful evaluation despite the time constraint. The diagnostic procedure predicted ‘facet_misweighting’ after inspecting the facet contribution event.

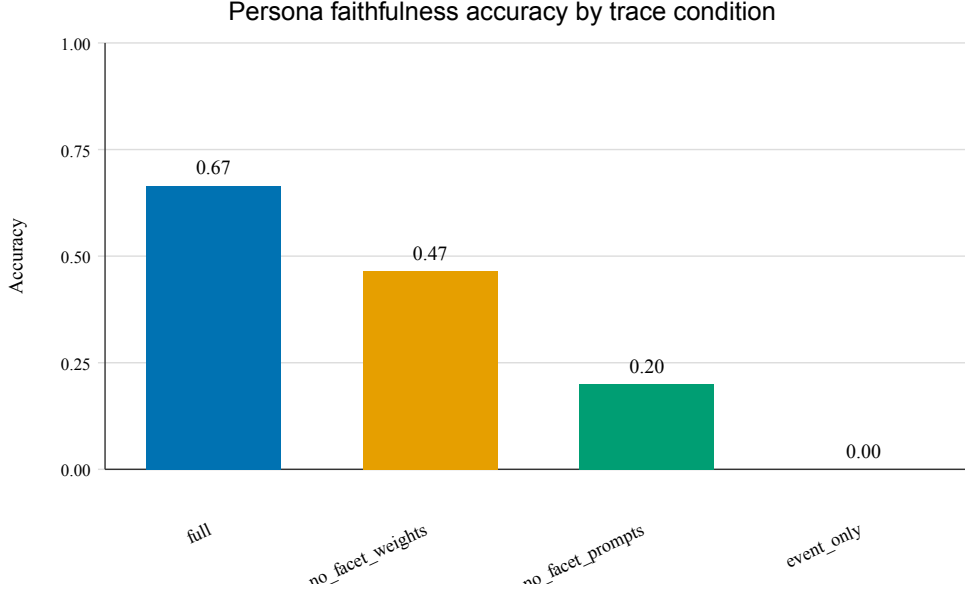


Figure 2: Facet-frame faithfulness diagnosis accuracy by trace condition.

10 Facet Faithfulness Audit Service Experiment

The benchmark above motivated a concrete implementation: **FacetFaithfulnessAuditService**. The service takes a persona response, selected facets, rendered prompt provenance, model metadata, self-reported confidence, and optional counterfactual probes. It returns a facet influence ledger, marker coverage, contradictions, counterfactual sensitivity, calibration score, final faithfulness score, primary diagnosis, causal facets, evidence, and recommended debug actions.

The service implements the audit hooks proposed by the framework: selected facet IDs and values, taxonomy versions, prompt hashes, configured and runtime weights, frame composition, suppressed facets, inferred interaction terms, expected markers, observed markers, forbidden marker contradictions, confidence calibration, and counterfactual sensitivity. A Vurvey adapter maps PM facet values, facet codes, **headerPrompt**, **detailPrompt**, confidence scores, taxonomy versions, and configured weights into this audit input shape.

We evaluated the audit service on eight hand-constructed Vurvey-style persona cases covering faithful responses, prompt omission, facet misweighting, facet conflict, persona drift, response-style violation, and low counterfactual sensitivity. The goal of this experiment is not to prove external predictive validity. It tests whether the implemented service can recover failure causes from increasingly rich evidence conditions.

Table 3: Facet faithfulness audit service results. Prompt-only and facet-list evidence are insufficient for reliable diagnosis. Adding weights improves diagnosis, and full AEF evidence, including prompt provenance and counterfactual probes, recovers all injected failures in this fixture suite. SNR is the ratio between target-facet intervention signal and paraphrase/noise sensitivity.

Condition	Accuracy	Faithfulness	Marker cov.	Counterfact.	Calibration	SNR
prompt_only	0.125	0.162	0.000	0.500	0.125	0.17
facet_list	0.125	0.382	0.000	0.500	0.125	0.83
facet_weights	0.750	0.526	0.281	0.500	0.406	2.20
full_aef	1.000	0.536	0.281	0.660	0.406	4.11

The experiment exposed an important design lesson. Mean faithfulness score is not the same as diagnostic accuracy. The full audit condition intentionally surfaces more missing markers and calibration penalties than the weaker conditions, so the mean score can remain conservative even while diagnosis improves. This is desirable for an audit system: it should not inflate trust simply because it sees more evidence. The useful output is the diagnosis and debug action, not a single high score.

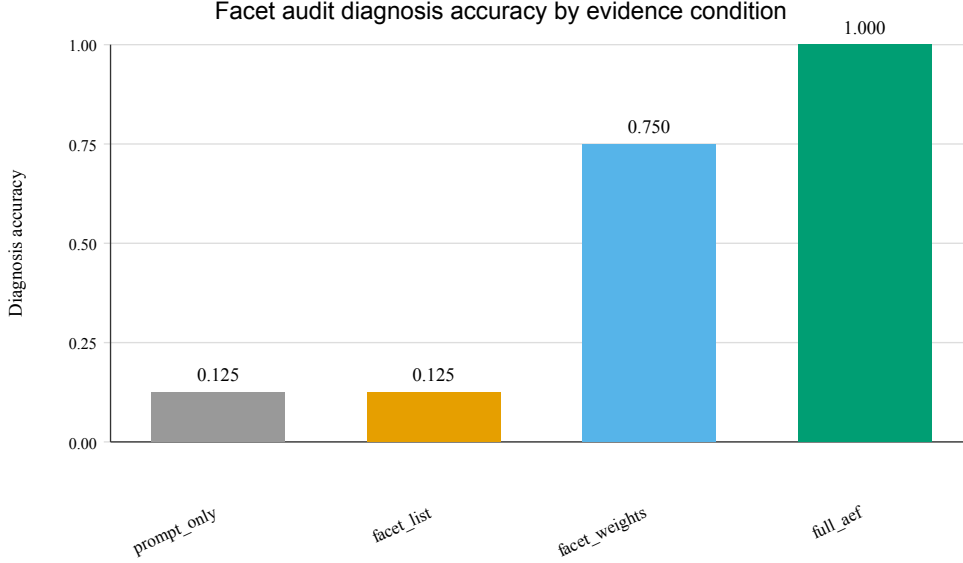


Figure 3: Facet audit diagnosis accuracy by available evidence condition.

In the faithful fixture, the ledger recorded five selected facets, prompt inclusion for each facet, prompt hashes, configured and runtime weights, and interaction terms such as `financial_outlook:crisis_flow x adoption_curve_position:innovator`. The service correctly marked the response as faithful while still listing missing secondary markers, such as explicit novelty language. In the prompt omission fixture, the rendered prompt omitted `financial_outlook`; the service diagnosed `facet_prompt_omission` and recommended verifying prompt assembly. In the persona drift fixture, the response introduced an unsupported enterprise CTO identity with unlimited budget; the service diagnosed `persona_drift` and recommended regenerating or constraining the persona biography.

11 Discussion

The generic benchmark shows that structured epistemic traces can recover simple injected failures when the trace contains the relevant fields. The facet benchmark is more realistic and harder. Full traces recover many failures, but not all. This is useful: persona faithfulness cannot be reduced to checking that a prompt contains the right words. It requires tracing facet selection, prompt rendering, frame composition, expected behavioral markers, response content, and sometimes counterfactual or latent evidence.

For Vurvey-style synthetic personas, AEF suggests several concrete audit hooks:

1. Record the selected facet IDs, value IDs, codes, taxonomy versions, and prompt hashes used for a response.
2. Record facet weights and interaction terms used to compose the active frame.
3. Record expected response markers derived from selected facets, such as cost objection, peer evidence request, time constraint, or response-style marker.
4. Run counterfactual probes that swap one high-impact facet while holding the rest constant.
5. Track self-rated confidence and compare it with evaluator judgments and response consistency.
6. When model internals are available, add representation-level probes for persona-relevant directions or sparse features.

12 Limitations

The benchmarks use controlled synthetic failures with known labels. They establish that the implementation can generate useful diagnostic traces, not that all synthetic personas are valid or predictive in

deployment. The facet benchmark uses deterministic heuristics for diagnosis and should be extended with noisy traces, human debugging studies, live product prompt paths, and held-out response data.

The framework also depends on the quality of facet definitions, prompt fragments, weights, and constraints. If the facet graph itself encodes weak theory or biased assumptions, AEF will make those assumptions more visible but will not make them correct. Detailed observer logs can expose sensitive persona or respondent data, so production use requires careful retention, access control, and redaction policy.

13 Code and Data Availability

The reference implementation is available at <https://github.com/jroell/agentive-epistemology-framework> under the MIT license. The implementation includes `FacetFaithfulnessAuditService`, the generic root-cause benchmark, the facet-frame benchmark, and the facet audit service experiment. Generated outputs are written under the repository’s results directory. A Vurvey API branch named `AEF-implementation` contains the product-facing adapter that maps PM facet values into the audit input shape.

14 Conclusion

The Agentive Epistemology Framework provides an audit architecture for autonomous agents and synthetic personas. By making beliefs, confidence, justifications, frames, facets, and observer events first-class constructs, AEF supports debugging not only what an agent said, but why the system should have expected that behavior from its configured epistemic state. The facet extension connects this audit layer to synthetic human simulation: facets become inspectable contributors to frame composition, and persona faithfulness becomes a measurable, diagnosable property rather than a matter of surface plausibility.

References

- Carlos E. Alchourrón, Peter Gärdenfors, and David Makinson. On the logic of theory change: Partial meet contraction and revision functions. *The Journal of Symbolic Logic*, 50(2):510–530, 1985.
- Michael E. Bratman. *Intention, Plans, and Practical Reason*. Harvard University Press, Cambridge, MA, 1987.
- Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. 2023. URL <https://arxiv.org/abs/2309.08600v3>.
- Phan Minh Dung. On the acceptability of arguments and its fundamental role in nonmonotonic reasoning, logic programming and n-person games. *Artificial Intelligence*, 77(2):321–357, 1995.
- Joshua M. Epstein and Robert Axtell. *Growing Artificial Societies: Social Science from the Bottom Up*. Brookings Institution Press, Washington, DC, 1996.
- Ronald Fagin, Joseph Y. Halpern, Yoram Moses, and Moshe Y. Vardi. *Reasoning About Knowledge*. MIT Press, Cambridge, MA, 2003.
- Charles J. Fillmore. Frame semantics and the nature of language. *Annals of the New York Academy of Sciences*, 280(1):20–32, 1976.
- Peter Gärdenfors. *Knowledge in Flux: Modeling the Dynamics of Epistemic States*. MIT Press, Cambridge, MA, 1988.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, Deep Ganguli, Danny Hernandez, Josh Jacobson, Jackson Kernion, Shauna Kravec, Liane Lovitt, Kamal Ndousse, Catherine Olsson, Sam Ringer, Dario Amodei, Tom Brown, Jack Clark, Nicholas Joseph, Ben Mann, Sam McCandlish, Chris Olah, and Jared Kaplan. Language models (mostly) know what they know. 2022. URL <https://arxiv.org/abs/2207.05221v4>.

- George Lakoff. *Women, Fire, and Dangerous Things: What Categories Reveal about the Mind*. University of Chicago Press, Chicago, IL, 1987.
- Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Jackson Kernion, Kamile Lukosiute, Karina Nguyen, Newton Cheng, Nicholas Joseph, Nicholas Schiefer, Oliver Rausch, Robin Larson, Sam McCandlish, Sandipan Kundu, Saurav Kadavath, Shannon Yang, Thomas Henighan, Timothy Maxwell, Timothy Telleen-Lawton, Tristan Hume, Zac Hatfield-Dodds, Jared Kaplan, Jan Brauner, Samuel R. Bowman, and Ethan Perez. Measuring faithfulness in chain-of-thought reasoning. 2023. URL <https://arxiv.org/abs/2307.13702v1>.
- Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models. 2023. URL <https://arxiv.org/abs/2303.08896v3>.
- Marvin Minsky. A framework for representing knowledge. Technical Report AI Memo 306, MIT AI Laboratory, Cambridge, MA, 1974.
- Donald Nute. Defeasible logic. *Handbook of Logic in Artificial Intelligence and Logic Programming*, 3: 353–395, 1994.
- Anand S. Rao and Michael P. Georgeff. Modeling rational agents within a BDI-architecture. In *Proceedings of the 2nd International Conference on Principles of Knowledge Representation and Reasoning*, pages 473–484. Morgan Kaufmann, 1991.
- Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting. 2023. URL <https://arxiv.org/abs/2305.04388v2>.
- Michael Wooldridge. *An Introduction to MultiAgent Systems*. John Wiley & Sons, Chichester, UK, 2nd edition, 2009.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to ai transparency. 2023. URL <https://arxiv.org/abs/2310.01405v4>.